Defense Method	TPR	TNR
Human-Crafted Prompt (Chiang et al., 2024)	0.64	**0.91**
Perplexity-based Detection (Alon and Kamfonas, 2023)	0.54	0.73
LLaMA 3.1 Guard (Inan et al., 2023)	0.61	0.81
Our Method (AEGIS @ Iteration 4)	0.76	0.88
Our Method (AEGIS @ Iteration 8)	**0.84**	0.89

Table 1: Defense Method Comparison. The result shows that our method already achieves the best TPR at iteration 4, despite a slight decrease in TNR compared with Human-Crafted Prompt. The model even performs better at iteration 8, with better TPR and TNR compared with our method at iteration 4.

achieving the best balance of a high True Positive Rate (**0.84**) and a high True Negative Rate (**0.89**), demonstrating its superior ability to identify sophisticated attacks while maintaining utility on benign inputs.

5.2 Iterative Performance

To illustrate the co-evolution of the attacker and defender, Table 2 shows the real-world evaluation for both agents at different iteration of the GAN training process. To be more specific, we evaluate the generated attacks against the defense in Human-Crafted Prompt from Chiang et al. (2024), and evaluate the generated defenses against the real-world articles (143 malicious articles + 100 benign articles) to calculate the TPR and TNR. The results demonstrate a clear trend of mutual improvement, where each agent becomes progressively stronger by adapting to the other.

Iteration	Attacker ASR	Attacker ΔS_{rel}	Defender TPR	Defender TNR
0	0.97	0.01	0.08	**0.99**
2	0.99	0.67	0.75	0.88
4	0.96	0.87	0.78	0.89
6	**1.00**	**1.00**	0.79	0.89
8	**1.00**	**1.00**	**0.84**	0.88

Table 2: Iterative Performance of Attacker and Defender evaluated on real world articles. The result shows that when the GAN iteration increases, both ASR and relative score change increases progressively. At the same time, TPR improves hugely despite a small decrease in TNP. This shows that the framework keeps finding better attacks and defenses that can perform well in the real world scenarios.

The trend shows the attacker's metrics (ASR, Relative Score Change) steadily increasing as it learns to bypass the improving defender. Simultaneously, the defender's TPR improves as it learns to find the defense prompt to detect the adversarial attacks. This dynamic demonstrates a successful adversarial training loop, leading to highly robust agents.

Appendix A.4 provides examples of real prompt refinements, illustrating that the defense yields meaningful qualitative improvements.

5.3 Cross-Model Generalizability

To evaluate the robustness and generalizability of the framework, we run the experiment on different LLMs.

5.3.1 Framework Generalizability

We first test whether our framework can be transfered on different LLMs, including GPT-5-mini, GPT-4.1-nano, Gemini-2.0-flash, Gemini-2.5-flash-lite. The results are shown in Table 3, We can see that all these models achieve better TPR at higher GAN iteration compared with lower GAN iteration, indicating that this framework can be applied on various LLMs. For more detailed results, please check the Appendix A.5.1.

GAN Iteration	GPT-5-mini	GPT-4.1-nano	Gemini-2.0-flash	Gemini-2.5-flash-lite
0	0.09	0.01	0.28	0.08
2	0.70	0.25	0.71	0.78
4	0.90	0.34	0.82	0.89

Table 3: Cross-Model Generalizability of Prompts Generated by GPT-4.1-mini. The values shown in the table is the True Positive Rate (TPR) for the generated defense prompt in GAN iteration 0, 2, and 4. All the prompts improve from iteration 0 to iteration 4 with different LLMs, meaning that the framework has great generalizabiliy.

5.3.2 Prompt Transferability

We also investigated the transferability of the generated defense prompts across various large-language models (LLMs). Specifically, we examine whether the prompts optimized using one model (in this case, GPT-4.1-mini) retain their effectiveness when applied to different LLMs (in this case, GPT-4.1-nano, Gemini-2.5-flash, and Gemini-2.5-flash-lite) without modification.

Table 4 illustrates the results for all models at GAN iteration 0, iteration 4, and iteration 8. For stronger LLMs (GPT-4.1-mini, Gemini-2.5-flash), they achieve a high TPR at iteration 8, while weaker LLMs (GPT-4.1-nano, Gemini-2.5-flash-lite) achieve lower TPR at iteration 8. However, we can see that the results for all LLMs improve when more GAN iterations are trained. For more detailed results, please check the Appendix A.5.2.

Gan Iteration	GPT-4.1-mini (source)	GPT-4.1-nano	Gemini-2.5-flash	Gemini-2.5-flash-lite
0	0.08	0.01	0.62	0.04
4	0.78	0.07	0.91	0.21
8	0.84	0.15	0.98	0.39

Table 4: Cross-Model Generalizability of Prompts Generated by GPT-4.1-mini. The values shown in the table is the True Positive Rate (TPR) for the generated defense prompt in GAN iteration 0, 4, and 8.

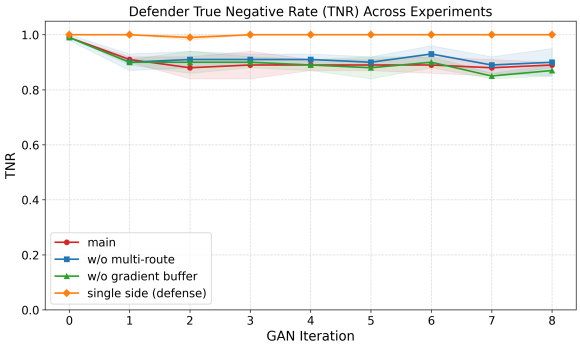

Figure 4: Iterative performance of the attacker and defender across GAN iterations, measured by True Negative Rate (TNR). Shaded regions represent the standard deviation across runs. No obvious difference can be seen for each ablation setup, all of them achieving TPR around 0.9 at each iteration.

6 Ablation Study

To understand the contribution of each component in AEGIS , we conducted several ablation studies. These studies involve systematically removing or altering parts of our system and observing the impact on performance.

6.1 Without Gradient Buffer

We removed the gradient buffer, which stores historical textual gradients for the optimization prompt. This change leads to a slower convergence rate and degrades the TPR in defense about 5%.

6.2 Without Multiple Gradients

We simplify the gradient generation process to use only one optimization type. For the defender, we used only the True Positive Rate (TPR), and for the attacker, only the Attack Success Rate (ASR). This led to a performance degrade in both attacker and defender, where the final performance of the defense degrades with over 10

6.3 Single-Sided Training

We also experimented with training only one side of the GAN framework. The lack of an adaptive adversary meant that the trained model quickly overfit to its static opponent, highlighting the importance

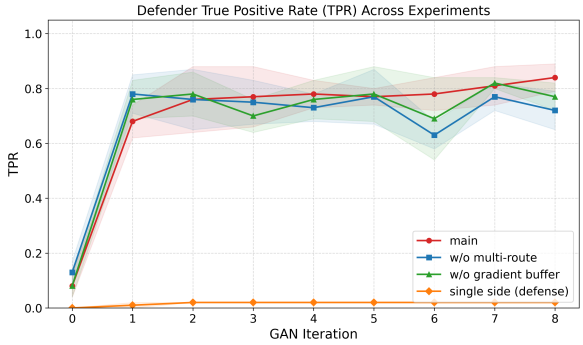

Figure 5: Iterative performance of the attacker and defender across GAN iterations, measured by True Positive Rate (TPR). Shaded region represent the standard deviation across runs. The default method has the steadiest improvement and achieve the best TPR at last.

of co-evolution. When training the defender against a static set of attacks, it fails to generalize to new, unseen attacks, resulting in a lower overall TPR.

The comparisons between all the ablation studies are presented in the figures above. Fig 5 compares the TPR between each ablation, and Fig 4 compares the TNR between each ablation.

7 Conclusion

We presented AEGIS , a novel automated co-evolutionary framework for guarding against prompt injection attacks in Large Language Models (LLMs). By iteratively optimizing both attack and defense prompts using a gradient-based natural language strategy, AEGIS systematically explores the prompt space without manual engineering. Our approach demonstrates superior performance over baseline and hand-crafted prompts on several LLMs, achieving promising results in both attack strength and defense robustness.

Through extensive experiments and ablation studies, we confirmed the importance of co-evolution, gradient replay, and multi-objective optimization. These findings suggest that adversarial training, when applied at the prompt level, offers a scalable and effective solution for safeguarding LLMs in real-world deployments. In future work, we plan to extend AEGIS to more complex scenarios and improve prompt interpretability.

Limitation

Despite the promising results, our study has several limitations. First, our evaluation focused on automated assignment grading task, which may not fully capture the diversity of real-world security-sensitive tasks and show the generalizability of our framework. Second, our defense method mainly targets text-based dialogue systems, and its effectiveness in multimodal systems remains unclear. Third, while we focused on quantitative evaluation of attack success rates and defense robustness, large-scale human evaluations were not conducted. Addressing these limitations in future work will be important for building more comprehensive and deployable defense systems.

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